Paul Brunzema

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My research lies at the intersection of **uncertainty quantification** (Gaussian processes, Bayesian neural networks) and **sequential decision-making** (Bayesian optimization, control, reinforcement learning). I aim to create agents that efficiently and optimally act in uncertain and potentially changing environments.

EDUCATION

RWTH AACHEN UNIVERSITY
Dr.-Ing. (PhD)
Institute for Data Science in Mechanical Engineering
Expected Mar 2027 | Germany
Supervisor: Prof. Dr. Sebastian Trimpe
Associated Researcher at RTG UnRAVeL

M.Sc. Automation Engineering
Dec 2021 | Germany
GPA: 1.1/1.0 (with distinction)
Top 5% | Dean's List

B.Sc. Mechanical Engineering
Mar 2019 | Germany
GPA: 2.2/1.0

UNIVERSITY OF CAMBRIDGE
ELLIS Summer School
Jul 2023 | UK
Probabilistic Machine Learning

GYMNASIUM KRONSHAGEN
Abitur
Jun 2015 | Germany
GPA: 1.9/1.0
DPG-Abitur Award (Physics)

LINKS

GitHub:// brunzema
LinkedIn:// paul-brunzema
Scholar:// Paul Brunzema

SKILLS

PROGRAMMING
Python (JAX, PyTorch, NumPy, ...)
Matlab • Shell • Git • ROS2
LaTeX • Simulink

LANGUAGES
German (native)
English (fluent)

INTERESTS & HOBBIES

Member of a democratic party in Germany
Support stem cell donor registration
Supervised F1-in-Schools team

Hobbies:
Chess • American football
CrossFit • Cycling • Board games
Hiking through beautiful nature

EXPERIENCE

RWTH AACHEN UNIVERSITY | Doctoral Researcher
03/01/2022 – 03/30/2027 (expected) | Theaterstr. 35-39, Aachen, Germany
- Research on (time-varying) Bayesian optimization and event-triggered learning
- Created MOOC on *Reinforcement Learning* (edX platform); in 2025, the course reached over 600 students across 83 countries.

META | Research Scientist Intern
05/18/2026 – 09/04/2026 | 770 Broadway, New York, NY
- Learning-based control for autonomous racing *(manuscript in preparation)*

TOYOTA RESEARCH INSTITUTE | Research Scientist Intern
06/23/2025 – 09/11/2025 | 4440 El Camino Real, Los Altos, CA
- Learning-based control for autonomous racing *(manuscript in preparation)*

RWTH AACHEN UNIVERSITY | Teaching & Graduate Assistant
05/01/2019 – 03/30/2022 | Templergraben 55, Aachen, Germany
- Assisted approx. 700 students in *Tutorial Control Engineering*
- Research on nonlinear model predictive control for fuel cell hybrid vehicles

PORSCHE ENGINEERING SERVICES | Intern
10/01/2018 – 03/30/2019 | Etzelstr. 1, Bietigheim-Bissingen, Germany

AWARDS & HONORS

2023	**SEW-EURODRIVE Student Award** Best Master's thesis, Faculty of Mechanical Engineering
2021-22	**Dean's List** (Top 5% in course of study)
2021	**MathWorks Fellowship** for Graduate Students
2021	**Germany Scholarship** by the RWTH Education Fund
2015	**DPG-Abitur Award** (Physics)

SELECTED PUBLICATIONS

Brunzema, P., et al. (2025). "Bayesian Optimization via Continual Variational Last Layer Training." *International Conference on Learning Representations (ICLR)*. **Spotlight**.

Brunzema, P., & Trimpe, S. (2025). "BayeSQP: Bayesian optimization through sequential quadratic programming." *Neural Information Processing Systems (NeurIPS)*. **Spotlight**.

Brunzema, P., et al. (2025). "Event-Triggered Time-Varying Bayesian Optimization." *Transactions on Machine Learning Research (TMLR)*.

Holzapfel, A.*, **Brunzema**, P.*, Trimpe, S. (2024). "Event-Triggered Safe Bayesian Optimization on Quadcopters." *Learning for Dynamics and Control (L4DC)*.

Brunzema, P.*, Kruse, P.*, Trimpe, S. (2024). "Neural Processes with Event Triggers for Fast Adaptation." *Learning for Dynamics and Control (L4DC)*.

Brunzema, P., et al. (2022). "On controller tuning with time-varying Bayesian optimization." *Conference on Decision and Control (CDC)*.

*Equal contribution — Full publication list at Google Scholar

Reviewing for: AAAI, NeurIPS, ICLR, ECML, CDC, ACC, L4DC; *Journals*: L-CSS, TCST, RA-L, Automatica